

Realizing a shell family by a faithful irreducible operator algebra

MathlibAnnex.CStarAlgebra.CAR.exists_completedAtomicShellModel

theorem

Constructs unitary links between pure-state summands and an irreducible algebra containing a faithful copy of CAR.

Statement

Let A be the completed CAR algebra and fix a shell family as defined below. Let $\mathcal{H} = \bigoplus_{j \in I} H_j$ be the Hilbert direct sum of the chosen pure-state Gelfand-Naimark-Segal (GNS) spaces, $\pi = \bigoplus_j \pi_j$ their representation, and $b_j = e_j \xi_j$ the embedded unit cyclic vectors, where $\xi_j \in H_j$ is the selected unit cyclic vector and $e_j : H_j \rightarrow \mathcal{H}$ is coordinate inclusion. There exist unitaries L_j on $\mathcal{H}$ with $L_j b_j = b_o$, $L_j \pi(\alpha_j(p_n)) = \pi(w_{j,n})$ for every j, n , and $L_o = I_{\mathcal{H}}$. The norm-closed unital star algebra $T = C^*(\pi(A), \{L_j\})$ in the bounded operators $B(\mathcal{H})$ contains an injective copy of A and acts irreducibly on $\mathcal{H}$.

Assumptions

Let A be the completed CAR algebra with matrix stages $M_{2^n}(\mathbb{C})$ embedded by $a \mapsto a \otimes I_2$. Write f_n for the image of the first diagonal matrix unit, $p_n = f_n - f_{n+1}$, and ω for the root state characterized by $\omega(\iota_n(a)) = a_{00}$ on each stage, where ι_n is its canonical embedding.

Index chosen pure-state Gelfand-Naimark-Segal (GNS) representations by their unitary-equivalence classes $j \in I$, with root index o and representative states φ_j .

A fixed shell family consists of complex-linear unital star automorphisms α_j and elements $w_{j,n} \in A$ satisfying $\varphi_j(\alpha_j(a)) = \omega(a)$, $w_{j,n}^* w_{j,n} = \alpha_j(p_n)$ and $w_{j,n} w_{j,n}^* = p_n$.

At the root, α_o is the identity and $w_{o,n} = p_n$. The Hilbert spaces are the chosen complete complex GNS spaces. No countability of I is assumed. Irreducibility means a nonzero representation with no proper nonzero closed reducing subspace.

Conclusion

The shell identities are realized by actual bounded unitary operators on one Hilbert space. The generated algebra acts irreducibly while retaining the original algebra faithfully; rank-one limiting projections and irreducibility are proved consequences, not fields imposed on the input family.

This is existence of the operator links for a supplied family. It does not construct that family without hypotheses, nor prove uniqueness of every irreducible representation of T . The universal capture theorem supplies that additional conclusion. No generic homogeneity principle is a further hypothesis of this exact theorem.

Proof route

By the atomic common-range theorem, the decreasing projections $\pi(\alpha_j(f_n))$ start at the identity and have common range $\mathbb{C}b_j$; the root flag has common range $\mathbb{C}b_o$. The represented links $\pi(w_{j,n})$ identify their corresponding orthogonal difference shells. The cited atomic-shell construction joins these partial isometries and the map between the one-dimensional residual spaces into unitaries. Its irreducibility argument uses the inequivalent irreducible GNS summands and the links between their cyclic vectors. Faithfulness comes from the faithful root summand of π .

▼ Proof steps

1. The support equations give initial projection $\pi(\alpha_j(f_n) - \alpha_j(f_{n+1}))$ and final projection $\pi(f_n - f_{n+1})$ for each represented shell link.
2. Use the proved common-range identity to identify the residual projections with the rank-one projections onto b_j and b_o . Apply the atomic-shell existence theorem, which supplies the unitaries and irreducibility of the generated algebra.
3. At the root the constructed link is the identity on every difference shell and on the residual line. These subspaces reconstruct $\mathcal{H}$, so $L_o = I_{\mathcal{H}}$. The faithful root summand makes π , hence its codomain restriction to T , injective.

Main citations

- Exact statement and proof
- A fixed family of compatible shell data
- Automorphisms and partial-isometry shell links
- Direct sum of the chosen pure GNS representations
- Faithfulness of the root GNS summand
- Faithfulness of the atomic source representation
- Common atomic range is the chosen cyclic line
- Initial support of a shell link
- Final support of a shell link
- Joining shells into an irreducible unitary model
- The faithful irreducible model captures every class
- The root state bundled with its purity proof

Lean source signature (exact)

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theorem exists_completedAtomicShellModel (family : RepresentativeShellFamily) :  
  ∃ L : MathlibAnnex.CStarAlgebra.PureState.GNSClass Limit →  
    MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert completedRootPureState
```

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→L[ $\mathbb{C}$ ]
    MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert completedRootPureState,
    (∀ i, L i ∈ unitary
      (MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert completedRootPureState
→L[ $\mathbb{C}$ ]
    MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert
completedRootPureState)) ∧
    (∀ i, L i (MathlibAnnex.CStarAlgebra.PureState.selectedEmbedding
completedRootPureState i
      (MathlibAnnex.CStarAlgebra.PureState.selectedVector completedRootPureState i)) =
      MathlibAnnex.CStarAlgebra.PureState.selectedEmbedding completedRootPureState
completedRootPureState.classOf
      (MathlibAnnex.CStarAlgebra.PureState.selectedVector completedRootPureState
completedRootPureState.classOf)) ∧
    (∀ i n, (L i).comp (selectedAtomicRepresentation
      (transportedFlag family i n - transportedFlag family i (n + 1))) =
      representedShellLink family i n) ∧
    L completedRootPureState.classOf = 1 ∧
    Function.Injective
      (MathlibAnnex.CStarAlgebra.AtomicConstruction.sourceHom
selectedAtomicRepresentation L) ∧
    Representation.IsIrreducible
      (MathlibAnnex.CStarAlgebra.AtomicConstruction.ambientInclusion
selectedAtomicRepresentation L)

```

In the source	Mathematical meaning
family : RepresentativeShellFamily	The supplied family $(\alpha_j, w_{j,n})$ for the completed CAR algebra A , root state ω , root index o , and projections $p_n = f_n - f_{n+1}$. It satisfies $\varphi_j \circ \alpha_j = \omega$, $w_{j,n}^* w_{j,n} = \alpha_j(p_n)$, $w_{j,n} w_{j,n}^* = p_n$, and the stated root normalization. Its existence is not this theorem's conclusion.
GNSClass Limit; SelectedAtomicHilbert completedRootPureState	The index set I of pure-GNS unitary-equivalence classes and the Hilbert sum $\mathcal{H} = \bigoplus_{j \in I} H_j$ of selected GNS spaces. The selected unit vectors are ξ_j , their coordinate inclusions e_j , and $b_j = e_j \xi_j$. No countability is imposed.
$\exists L : \dots \rightarrow \dots \rightarrow L[\mathbb{C}] \dots$	There is one family of bounded complex-linear operators $L_j : \mathcal{H} \rightarrow \mathcal{H}$ satisfying all six following clauses simultaneously.
$\forall i, L i \in \text{unitary } (\dots)$	First, every L_j is unitary on $\mathcal{H}$: $L_j^* L_j = L_j L_j^* = I$.
$\forall i, L i (\text{selectedEmbedding } \dots i (\text{selectedVector } \dots i)) = \dots$	Second, each same L_j sends $b_j = e_j \xi_j$ to the root vector $b_o = e_o \xi_o$; the fully qualified names in the signature select exactly these vectors and inclusions.

In the source	Mathematical meaning
$\forall i \in n, (L_i).comp (selectedAtomicRepresentation (transportedFlag\ family\ i\ n - transportedFlag\ family\ i\ (n + 1))) = representedShellLink\ family\ i\ n$	Third, for every j, n, $L_j \pi(\alpha_j(f_n) - \alpha_j(f_{n+1})) = \pi(w_{j,n})$. Here $\pi = \bigoplus_j \pi_j$, <code>transportedFlag</code> is $\alpha_j(f_n)$, and <code>.comp</code> applies the rightmost operator first.
<code>L.completedRootPureState.classOf = 1</code>	Fourth, the root member is $L_o = I_{\mathcal{H}}$.
<code>Function.Injective (AtomicConstruction.sourceHom selectedAtomicRepresentation L)</code>	Fifth, the codomain restriction $\iota : A \rightarrow T$, $\iota(a) = \pi(a)$, is injective, for $T = C^*(\pi(A), \{L_j : j \in I\})$ built from this same L.
<code>Representation.IsIrreducible (AtomicConstruction.ambientInclusion selectedAtomicRepresentation L)</code>	Sixth, the inclusion representation $\rho : T \rightarrow B(\mathcal{H})$ of that same generated algebra is nonzero and has no proper nonzero closed reducing subspace. This is distinct from injectivity of ι.

Further source notes: Here `Limit` is A and `completedRootPureState` is the root state ω , bundled with its purity proof. It does not assert existence of the input family.

Earlier published Card and PDF

▼ Exact content identity

Declaration: `MathlibAnnex.CStarAlgebra.CAR.exists_completedAtomicShellModel`
 Accepted content SHA-256: `b3c1c2a0e02b60f44fca567e146983edcace16a3b6335e6d9a480c3e4c5709b`
 Accepted source guide SHA-256: `c1b8891ddbdfcf047307b402e9786f0c45d608a23e6b1b34592e4149ce7d68d5`
 Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`